

Hovercraft

(Coach Scheduled Event - 45 Minutes)



Description: Teams will design, construct, and test a balloon hovercraft that levitates on a cushion of air. Choosing their desired ramp and starting position on the ramp, competitors will send their hovercraft down the ramp to stop as close to a finish line as possible, without passing the finish line. Competitors will also complete a written assessment covering hovercraft design, potential and kinetic energy, and friction.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Team members must also bring:

- Pencils
- Scissors
- Teams may bring a straight edge measuring device
- Students may bring a log of previous hovercraft tests to refer to during their device testing, if desired. No additional notes beyond such a log may be brought into the competition.

Materials Provided at the Event:

- 1 - standard size DVD/CD (12 cm)
- 1 - 20 cm (8 in) balloon
- 7.5 ml to 15ml (1/4 to 1/2 oz) mounting putty
- 1 - 2.7 cm (1.1 in) diameter plastic soda bottle pop top
- 45 cm (18 in) of 2.54 cm (1 in) masking tape
- 1 - 8 ½ in x 11 inch piece of tracing paper
- 1 - 8.5 X 11-inch 60–90-pound card stock paper
- Test Track - see below for parameters

THE COMPETITION

The Written Assessment: The team members will complete an assessment on hovercraft design, potential and kinetic energy, and friction. The assessment will be no longer than 20 questions and will take place simultaneously during the construction of the team hovercraft.

Construction Phase (15 min):

- The team may use any design to create their hovercraft using any or all of the provided materials.

- Teams may request a second balloon if they break their first one, but will not receive any additional beyond that.

The Track:

- The track is in two parts - the ramp and the portion on the floor.
- The ramp will be approximately 20 cm (8 in) wide, 122 cm (48 in) long, and approximately 14 cm (5.5 in) tall at the top, with guide rails measuring 2 cm ($\frac{3}{4}$ in) tall along the sides. The surface of the ramp will be the smooth side of a hardboard wood wall panel (also called MDF).
- Two lengths of wood, 1" x 2" in cross section, will be placed on the floor aligned with the guide rails of the ramp to extend the track onto the floor. These lengths are intended to keep the hovercraft contained when on the floor, and should extend at least 6 feet beyond the end of the ramp.
- A finish line will be marked on the floor side rails. The finish line will be anywhere from 0.5 meters - 1.5 meters from the end of the ramp section.

Hovercraft testing

- The goal of the hovercraft trials is to build the hovercraft and select a starting position on the ramp so that the team's hovercraft travels as close to the finish line as possible, while not crossing it.
- Teams will get two runs and the better of their runs will be used in the event scoring.
- Teams will decide where on the ramp to hold their balloon hovercraft and hold it so that no air escapes. When the Event Supervisor says that they are ready, the team will release the hovercraft on the track.
- When the hovercraft stops, the Event Supervisor will record the distance in cm from the front edge of the hovercraft to the finish line and whether the hovercraft crossed the finish line.
- The team will receive no score for the run if they touch the hovercraft during the run or pick it up before the Event Supervisor can measure its distance.
- The Event Supervisor will not allow teams to compete with their hovercraft if they deem the team's device to be unsafe in any way.

SCORING: Final Score (FS) = Assessment Score + Track Score, max score = 100 points

- Assessment score = $(TAS \div CAS) \times 60$; rounded to the 100th place value; maximum 60 points
 - Team Assessment Score (TAS) = team's score on the assessment
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Track Score = $(CTS \div TTS) \times 40$; rounded to the 100th place value; maximum 40 points
 - Distance Score = Crossing Multiplier x ES-measured distance
 - If the hovercraft finishes before the finish line, the Crossing Multiplier = 1
 - If the hovercraft crossed the finish line, the Crossing Multiplier = 2
 - Competition Track Score (CTS) = the lowest Distance Score for all teams
 - Team Track Score (TTS) = the team's best Distance Score

Scoring Example:

- Competition Assessment Score (CAS) = 15. The team scores 9 on the assessment.
Assessment Score for the team = $(9 \div 15) \times 60 = 36$ points
- On the team's first run, the hovercraft crossed the finish line and was 3 cm from it.
Distance Score on Run 1 = $2 \times 3 = 6$
On the team's second run, the hovercraft finished 10 cm in front of the finish line.
Distance Score on Run 2 = $1 \times 10 = 10$
So Team's Track Score = 6 (the better of the two)
- Competition Track Score (CTS) = 3 (the best of all team's runs)
Track Score for the team = $(3 \div 6) \times 40 = 20$ points
- Final Score (FS) = $36 + 20 = 56$ points

Tiebreakers:

- 1st - Best Team Track Score (TTS)
- 2nd - Best Team Assessment Score (TAS)

Resources:

- [NASA JPL Education — Make a Hovercraft](#)
- [Go Science Girls — Balloon Hovercraft](#)
- [Sublime Science — CD Hovercraft](#)
- [KiwiCo — Balloon Hovercraft](#)
- [The Science Magician — Balloon Hovercraft](#)
- [Fab Science — Balloon-Powered Hovercraft](#)
- [Science Bob — Tabletop Hovercraft](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles